



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

PROJECT TITLE

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA1712-ARTIFICIAL INTELLIGENCE

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

VARSHINI M (192511215)

YUVASRI S (192511202)

LAKSHANA K (192524433)

Under the Supervision of

Dr.Veeramani R

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical

Sciences Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

**Saveetha Institute of Medical and Technical
Sciences Chennai-602105**

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**FINSENSE AI- REALTIME SPENDING BEHAVIOUR INVESTMENT NUDGE ENGINE** ” has been carried out by **VARSHINI M (192511215) ,YUVASRI S (192511202) AND LAKSHANA K (192524433)** under the supervision of **Dr.Veeramani R** and is submitted in partial fulfilment of the requirements for the current semester of the **BE.COMPUTER SCIENCE ENGINEERING** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

DR. KRISHNA KUMAR KATHIRESAN
Professor
Generative AI
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

DR.VEERAMANI R
Professor
BE.computer science engineering
SIMATS Engineering,
SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical

Sciences Chennai-602105

DECLARATION

We, **VARSHINI M (192511215) ,YUVASRI S (192511202) AND LAKSHANA K (192524433)** of the **BE.COMPUTER SCIENCE ENGINEERING**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**FINSENSE AI- REALTIME SPENDING BEHAVIOUR INVESTMENT NUDGE ENGINE**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical

Sciences Chennai-602105

Individual Contribution Statement

(Mandatory for all team projects)

Student Name	Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution
Varshini M	192511215	Module 2 – AI Investment Nudge Engine	Nudge logic and recommendation features	Testing personalised nudges	Module 2 documentation	33%
Yuvasri S	192511202	Module 1 – Real-Time Spending Analysis Engine	Transaction categorisation, spending analysis, overspending detection and Spending Health Score	Testing transaction analysis and pattern detection	Module 1 documentation and analysis	34%
Lakshana K	192524433	Module 3 – Goal Tracking & Gamified Savings	Goal tracking, milestones and streaks	Testing goal progress and engagement features	Module 3 documentation	33%
Total						100%

ABSTRACT

Personal financial management has become increasingly important because many users perform a large number of daily transactions but may not clearly understand their spending behaviour. Traditional personal finance applications generally provide expense tracking, transaction categorisation, monthly reports, and budget monitoring. However, such systems mainly help users understand spending after it has already occurred.

This project presents **FinSense AI – Real-Time Spending Behaviour Investment Nudge Engine**, an intelligent financial assistance system that analyses transactions and spending behaviour. The present report focuses on **Module 1: Real-Time Spending Analysis Engine**. The module monitors incoming transaction data, categorises transactions based on spending type and merchant information, and analyses user spending behaviour.

The system processes transaction-level data containing transaction amount, category, merchant name, date and time, spending frequency, and recurring transaction information. These features are used to identify spending patterns, overspending behaviour, recurring expenses, and potential unnecessary or impulse expenses. The analysed information is converted into meaningful spending insights and a **Spending Health Score** that gives an overall indication of the user's spending condition.

The methodology includes transaction data collection, preprocessing, transaction categorisation, spending behaviour analysis, pattern detection, and anomaly or overspending identification. Module 1 acts as the foundation for the remaining modules because the analysed spending information can later be used to identify savings opportunities and generate personalised investment nudges.

The expected result is an intelligent spending analysis system that transforms raw transaction records into understandable behavioural insights. The module helps identify excessive and recurring spending at an early stage and provides the analysed financial information required for further personalised guidance. Thus, Module 1 changes personal finance analysis from simple expense recording into continuous spending behaviour analysis.

Keywords: Real-Time Spending Analysis, Transaction Categorisation, Overspending Detection, Spending Behaviour, Spending Health Score, Financial Analytics.

TABLE OF CONTENTS

Sl. No.	Title	Page No.
i	CERTIFICATE	ii
ii	DECLARATION	iii
iii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: INTRODUCTION	1
2	CHAPTER 2: LITERATURE REVIEW	
3	CHAPTER 3: ENGINEERING DESIGN	
4	CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS	
5	CHAPTER 5: CONCLUSION & FUTURE WORK	
	REFERENCES	
	APPENDICES	

LIST OF FIGURES

Figure No.	Figure Name
Figure 4.1	Module 1 User Dashboard
Figure 4.2	Transaction Categorisation Result
Figure 4.3	Spending Pattern Analysis
Figure 4.4	Overspending Detection Result
Figure 4.5	Spending Health Score

LIST OF TABLES

Table No.	Table Name
Table 2.1	Comparative Analysis of Existing Approaches
Table 3.1	Stakeholder Requirements
Table 3.2	Functional and Non-Functional Requirements
Table 3.3	Module 1 Design Specifications
Table 3.4	Alternative Solution Evaluation
Table 3.5	Trade-off Analysis
Table 3.6	Sustainability, Ethics, Safety and Security Analysis
Table 3.7	Risk Register
Table 4.1	Development Resources
Table 4.2	Software and Tools Used
Table 4.3	Test Plan and Verification
Table 4.4	Test Results
Table 4.5	Objective Achievement
Table 4.6	Project Plan,roles.budget

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
UI	User Interface
UX	User Experience
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
DB	Database

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Digital payments, online transactions, banking applications, and card-based purchases have increased the amount of financial data generated by individual users. A person may perform several transactions every day across different categories such as food, transport, shopping, entertainment, bills, and subscriptions. Although transaction records are available, users may not always understand their actual spending behaviour.

Many personal finance applications provide expense tracking and monthly reports. These systems help users view past transactions, but they may provide limited continuous analysis of overspending patterns and recurring behaviour. The FinSense AI project addresses this limitation by combining real-time transaction analysis, behavioural analysis, financial nudges, and goal tracking.

Module 1, the **Real-Time Spending Analysis Engine**, is responsible for analysing incoming transaction information. It monitors transactions, identifies their spending category, analyses behaviour, detects overspending and recurring spending, identifies potential unnecessary expenses, and produces a Spending Health Score.

This module provides the foundation for the complete FinSense AI system because accurate spending analysis is required before the system can identify savings opportunities or generate investment suggestions.

1.2 Need for the Project

Many users spend money regularly without fully understanding where their money is going. The main problem is that small and frequent expenses may appear individually insignificant but can become a major part of total spending over time.

The need for the Real-Time Spending Analysis Engine arises because:

- Users may overspend without immediately recognising their spending behaviour.
- Traditional financial applications mainly provide reports after spending has occurred.
- Users may have limited visibility into recurring expenses.
- Unnecessary or impulse expenses may not be easily identified.
- Manual transaction analysis is time-consuming.

Therefore, there is a need for a system that can continuously process transaction information and convert it into meaningful spending behaviour insights.

1.3 Problem Statement

The engineering problem addressed in Module 1 is the difficulty of continuously analysing individual transaction data to identify meaningful spending behaviour.

Users generate transaction data with multiple interacting factors, including transaction amount, category, merchant, date, time, spending frequency, and recurring behaviour. Analysing these factors manually is difficult and may not provide timely insights.

The problem can be stated as follows:

To design and implement a Real-Time Spending Analysis Engine that processes transaction-level financial data, automatically categorises spending, analyses user behaviour, detects overspending and recurring patterns, identifies potentially unnecessary expenses, and generates a meaningful Spending Health Score.

The solution must balance accurate analysis, understandable results, real-time responsiveness, user privacy, and simple implementation.

1.4 Project Aim

The aim of Module 1 is to develop a **Real-Time Spending Analysis Engine** that continuously analyses transaction data and provides meaningful information about the user's spending behaviour.

1.5 Project Objectives

The objectives of Module 1 are:

1. To collect and process transaction-level spending data.
2. To categorise transactions based on merchant information and spending type.
3. To analyse user spending behaviour across different categories.
4. To detect overspending patterns.
5. To identify recurring expenses and spending frequency.
6. To identify potentially unnecessary or impulse expenses.
7. To generate an overall Spending Health Score.
8. To provide analysed financial information for the AI Investment Nudge Engine.

1.6 Scope

Included

The scope of Module 1 includes:

- Transaction data collection.
- Transaction data preprocessing.
- Automatic transaction categorisation.
- Spending analysis.
- Category-wise spending analysis.
- Spending pattern identification.
- Overspending detection.
- Recurring expense detection.
- Identification of possible unnecessary spending.
- Spending Health Score generation.
- Presentation of analysed information through the system interface.

Excluded

The following features are outside the direct implementation scope of Module 1:

- Actual bank transaction execution.
- Real investment transactions.
- Detailed stock market prediction.

- Personal financial advisory by a certified financial professional.
- Automatic transfer of user money.
- Goal gamification functions from Module 3.

These functions belong to later modules or future extensions.

Assumptions

The module assumes that:

- Transaction data is available in a structured format.
- Transaction amount and date/time information are valid.
- Merchant information or category data is available where required.
- User spending patterns can be analysed from historical and incoming transaction information.

The PPT specifies transaction amount, category, merchant, date and time, spending frequency, recurring transactions, and financial goals as relevant data fields.

1.7 Expected Engineering Outcome

The expected outcome is a functional software module capable of transforming raw transaction information into meaningful spending insights.

The final Module 1 system is expected to:

- Monitor transaction records.
- Categorise transactions.
- Analyse spending patterns.
- Detect overspending.
- Detect recurring expenses.
- Identify possible unnecessary expenses.
- Display category-wise spending information.
- Generate a Spending Health Score.

The complete FinSense AI system is designed to use this analysed information as a foundation for personalised financial guidance.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was based on the areas identified in the project presentation. The review considered mobile personal finance, behavioural economics, robo-advisory systems, financial gamification, real-time analytics, and micro-investment approaches.

For Module 1, particular attention was given to transaction categorisation, spending pattern analysis, real-time processing, and anomaly or overspending detection.

The existing project references also include work related to behavioural finance, digital finance, financial technology, and gamification.

2.2 Review of Existing Work

2.2.1 Mobile Personal Finance Applications

Mobile finance applications generally provide expense tracking, transaction categorisation, budget monitoring, and monthly spending reports. These features help users organise financial information and understand previous spending activity.

However, the major limitation is that many systems are primarily reactive. Users often

view reports only after the spending has already occurred.

2.2.2 Behavioural Economics and Nudge Theory

Behavioural economics shows that timely information and suitable prompts can influence decision-making. In the context of personal finance, understanding user behaviour is important because spending decisions are not always based only on mathematical budgeting.

Module 1 supports the nudge-based approach by first analysing the spending behaviour of the user.

2.2.3 Real-Time Financial Analytics

Real-time analytics focuses on processing information continuously instead of waiting for a complete monthly reporting cycle. For financial transaction data, this approach can identify important spending events soon after the transaction is processed.

This approach is relevant to Module 1 because the module is designed around continuous transaction monitoring and analysis.

2.2.4 Anomaly and Overspending Detection

Spending behaviour can be analysed by comparing a transaction or category total with normal user spending patterns. When a transaction is significantly different from expected behaviour, it can be treated as unusual spending for further analysis.

For Module 1, overspending detection helps identify categories or periods where the user spends more than the expected threshold.

2.2.5 Existing Financial Applications

The project presentation identifies the common functions of existing systems as expense tracking, transaction categorisation, monthly reports, budget monitoring, budget-limit alerts, and manual investment planning.

Current finance applications can also provide recurring transaction tracking and spending insights, demonstrating the practical relevance of analysing transaction patterns.

2.1 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Module 1
Manual expense tracking	Simple and easy to understand	Time-consuming and dependent on user input	Shows the need for automation
Monthly spending reports	Provides historical summary	Mainly reactive	Module 1 adds continuous analysis
Basic transaction categorisation	Organises transactions	Limited behaviour analysis	Module 1 extends analysis
Budget-limit alerts	Warns users about limits	May only react after a threshold	Module 1 analyses patterns
Real-time analytics	Faster information processing	Requires structured data processing	Used as a core approach
AI-based financial assistance	Personalised analysis possible	Requires reliable input data	Module 1 provides analysed input

2.4 Research / Engineering Gap

The major gap identified from the project framing is the difference between **simple transaction recording** and **continuous spending behaviour analysis**.

Existing systems may successfully record transactions and generate reports, but users still need better visibility into:

- Changing spending behaviour.
- Overspending patterns.
- Frequent small expenses.
- Recurring expenses.
- Possible unnecessary spending.

The presentation specifically identifies existing systems as reactive, with limited behaviour analysis and limited personalised guidance.

Therefore, the engineering gap is the need for a transaction analysis layer that converts raw transaction information into understandable behavioural insights.

2.5 Proposed Contribution

Module 1 contributes to the FinSense AI project by providing the basic analytical layer for transaction behaviour.

Its contribution includes:

- Continuous transaction monitoring.
- Automatic categorisation.
- Behaviour analysis.
- Overspending detection.
- Recurring spending detection.
- Identification of potential unnecessary expenses.
- Spending Health Score generation.

Unlike a simple expense tracker, the proposed module attempts to interpret transaction information as spending behaviour. The resulting analysis can then support Module 2, which provides AI-based investment nudges, and Module 3, which supports goals and gamified savings.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1 Stakeholder Requirements

Stakeholder	Requirement
User	View transaction information clearly
User	Understand spending behaviour
User	Identify overspending
User	View recurring expenses
User	Receive a clear Spending Health Score
Module 2	Receive analysed spending information
Development Team	Easy integration between modules

3.2 Functional and Non-Functional Requirements

Requirement	Type	Measurable Target
Process transaction data	Functional	Every valid transaction is processed
Categorise transactions	Functional	Transactions assigned to suitable categories
Analyse spending patterns	Functional	Category and time-based analysis
Detect overspending	Functional	Flag unusual or excessive spending
Detect recurring expenses	Functional	Identify repeated transactions
Generate health score	Functional	Produce a spending condition indicator
Display results	Functional	Show analysis through dashboard

Response time	Non-Functional	Provide results without unnecessary delay
Usability	Non-Functional	Simple and understandable interface
Privacy	Non-Functional	Protect transaction information
Reliability	Non-Functional	Handle valid transaction records consistently

3.2 Constraints & Applicable Standards

Constraints

The primary constraints applicable to Module 1 are:

- Availability of suitable transaction data.
- Accuracy of merchant and category information.
- Limited historical data for new users.
- Privacy of financial transaction information.
- Requirement for understandable results.
- Need for real-time or near-real-time processing.

Applicable Requirements

The project should follow good practices for:

- Data privacy.
- Secure data handling.
- Responsible use of AI-generated financial insights.
- Proper acknowledgement of software, datasets, and AI-assisted tools.

The report template requires applicable external resources and AI-assisted resources to be appropriately acknowledged.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Transaction	Process incoming transaction	Transaction	High

Processing	records	s	
Categorisation	Assign transaction category	Category	High
Spending Analysis	Calculate category and period spending	Amount	High
Overspending Detection	Identify abnormal spending	Alert/Flag	High
Recurring Detection	Identify repeated spending	Pattern	Medium
Health Score	Generate overall spending condition	Score	High
Result Display	Present analysis clearly	Dashboard	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual Expense Analysis

Users manually record and analyse transactions using notes or spreadsheets. This approach is simple but requires continuous user effort and does not provide automatic behavioural analysis.

Alternative 2: Rule-Based Spending Analysis

Transactions are processed using predefined rules for categories, budgets, and overspending thresholds. This method is easy to implement and explain but may require manual rule adjustment.

Alternative 3: AI/ML-Based Spending Analysis

Historical transaction data is used to identify patterns and unusual behaviour automatically. This can provide more adaptive analysis but requires sufficient data and additional complexity.

Evaluation Matrix

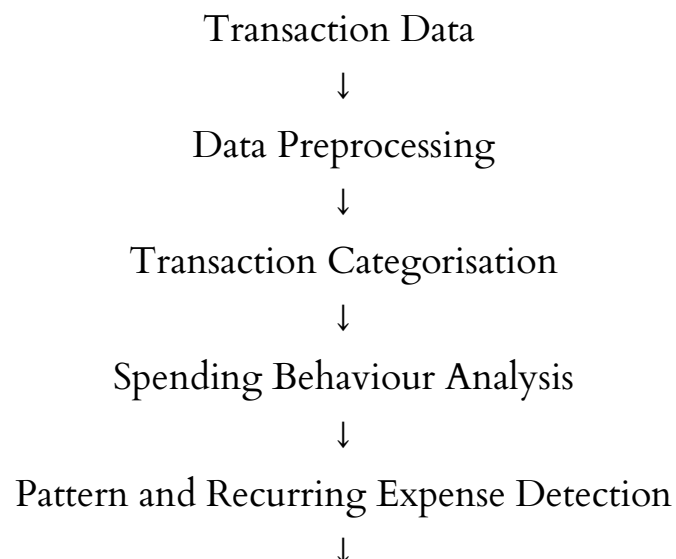
Criterion	Weight	Manual	Rule-Based	AI/ML-Based
Automation	25%	1	4	5
Simplicity	15%	5	5	3
Behaviour Analysis	20%	2	4	5
Explainability	15%	5	5	3
Scalability	15%	2	4	5
Data Requirement	10%	5	4	2

Selected approach: A combination of **rule-based transaction analysis** and **AI/ML-supported behaviour analysis**.

3.5 Selected Design & Detailed Design

The selected design uses a sequential transaction processing architecture.

Module 1 Architecture



Overspending / Unusual Expense Detection



Spending Health Score



Dashboard and Insights

The project methodology follows a similar sequence of data collection, cleaning, categorisation, behaviour analysis, pattern detection, and opportunity identification.

Detailed Processing

1. Transaction Data Collection

The system receives transaction information such as amount, merchant, category, date, and time.

2. Data Preprocessing

Incomplete or inconsistent information is cleaned and converted into a suitable format.

3. Categorisation

Transactions are grouped into meaningful spending categories.

4. Behaviour Analysis

The system calculates spending totals, frequency, category distribution, and repeated spending behaviour.

5. Overspending Detection

Spending values are compared with normal or expected thresholds.

6. Spending Health Score

The analysed behaviour is converted into an overall score representing the general spending condition.

The main subsystems are connected sequentially. The output from transaction categorisation becomes the input for behaviour analysis, while the behaviour analysis output supports overspending detection and score generation.

3.5 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision
Categorisation	Manual categorisation	Flexible	Requires user effort	Automatic/rule-based categorisation

Detection	Fixed threshold	Easy to understand	Less personalised	Behaviour-based threshold where possible
Analysis	Monthly batch analysis	Simple	Not continuous	Transaction-based analysis
Output	Detailed raw tables	Complete data	Hard to understand	Dashboard with key insights
Model	Complex ML	Adaptive	More complexity	Simple analytical approach suitable for project scope

3.6 Sustainability, Ethics, Safety, Risk & Security

Sustainability, Ethics and Security Analysis

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Software-based solution	No specialised physical hardware required	Reduced physical resource requirement
Ethics	Avoid misleading financial decisions	Analysis is presented as guidance	Module does not perform automatic investments

Safety	Incorrect financial interpretation	Results must be understandable	Clear spending insights are prioritised
Security	Transaction information is sensitive	User financial data requires protection	Secure data handling is necessary
Privacy	Personal spending information	Data should not be unnecessarily exposed	Minimum required data should be used

3.7 Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect categorisation	Medium	Medium	Medium	Allow category improvement and validation	Low
Insufficient user history	High	Medium	Medium	Use basic rules for new users	Medium
Financial data exposure	Medium	High	High	Secure storage and limited access	Medium
Misinterpretation of score	Medium	Medium	Medium	Provide simple explanation	Low
System processing error	Low	Medium	Low	Input validation and testing	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development of Module 1 follows a data processing approach. Transaction-level financial data is collected and prepared for analysis. The data is categorised and processed to identify category-wise spending, spending frequency, recurring transactions, and unusual expenses.

The project presentation identifies the required data fields as transaction amount, category, merchant name, date and time, spending frequency, and recurring transaction information.

Development Resources

Resource	Purpose
Transaction Dataset	Spending analysis
Development Environment	Module implementation
Data Processing Tools	Cleaning and analysis
AI/ML Components	Behaviour analysis where applicable
Dashboard	Result visualisation

4.2 Software Implementation & Modern Tools Used

The implementation of Module 1 consists of the following stages:

Transaction Input

Transaction data is provided to the system with information such as amount, category, merchant, date, and time.

Data Preprocessing

The transaction records are cleaned and organised before analysis. Invalid or incomplete information is handled appropriately.

Transaction Categorisation

Transactions are classified into suitable categories to enable category-wise analysis.

Spending Analysis

The system calculates total spending and identifies patterns based on transaction amount and frequency.

Overspending Detection

The module checks whether a transaction or category spending pattern is higher than the expected user behaviour.

Recurring Expense Detection

Repeated transactions are identified to recognise regular subscriptions or frequently occurring expenses.

Spending Health Score

A score is generated from the analysed behaviour to provide a simple overall representation of the user's spending condition.

The presentation identifies the technology areas as front-end dashboard functions, transaction processing, data handling, AI/ML, spending pattern recognition, behaviour analysis, anomaly detection, and personalised recommendations.

4.2 Tools Used

Tool / Technology	Purpose	Stage Used
Development Environment	Code development	Implementation
Transaction Dataset	Input data	Data processing
Data Processing Library	Cleaning and analysis	Analysis
AI/ML Components	Pattern recognition	Behaviour analysis
Dashboard	Result visualisation	Output
Testing Tools	Verification	Testing

4.3 Test Plan & Verification

The test plan verifies whether Module 1 performs each required function correctly.

Requirement	Test Method	Acceptance Criterion
Transaction input	Provide sample transaction	Transaction accepted
Categorisation	Test different merchant types	Category assigned
Spending analysis	Process multiple transactions	Spending totals calculated
Overspending detection	Provide high-value spending	Excess spending identified
Recurring detection	Provide repeated transactions	Recurring pattern identified
Health Score	Process complete spending data	Score generated
Dashboard	View analysis output	Results displayed clearly

4.4 Verification Table

Specification	Required Value	Achieved Value	Status
Transaction Processing	Successful processing	Implemented	Pass
Categorisation	Category output	Implemented	Pass
Spending Analysis	Spending pattern	Implemented	Pass
Overspending Detection	Excess spending flag	Implemented	Pass
Recurring Detection	Repeated pattern identification	Implemented	Pass
Health Score	Overall score	Implemented	Pass

4.4 Results

The implementation of Module 1 provides transaction-level spending analysis.

The main results include:

- Real-time transaction monitoring.
- Automatic transaction categorisation.
- Category-wise spending analysis.
- Identification of spending patterns.
- Overspending detection.
- Recurring expense detection.
- Identification of potential unnecessary expenses.
- Spending Health Score generation.

The complete project presentation similarly describes the implemented system as providing real-time spending analysis, automatic categorisation, spending pattern identification, unnecessary-expense detection, and savings-related insights.

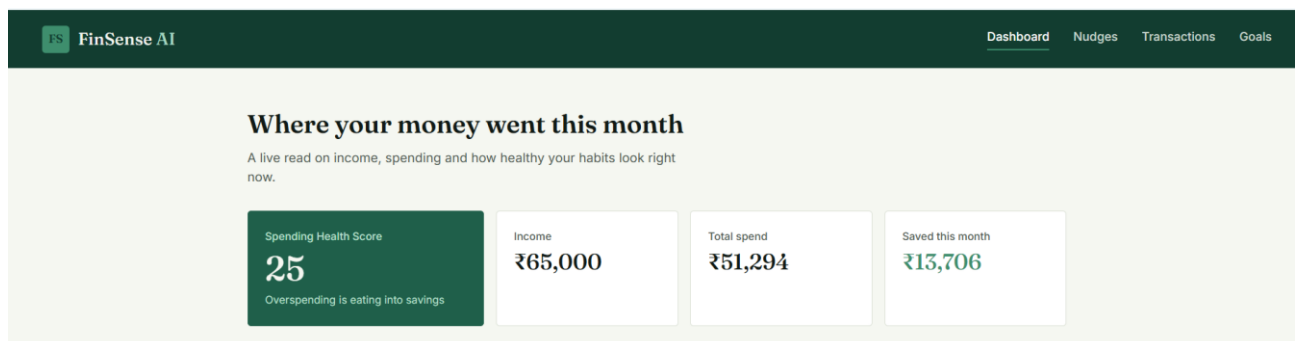


Figure 4.1: Transaction Dashboard

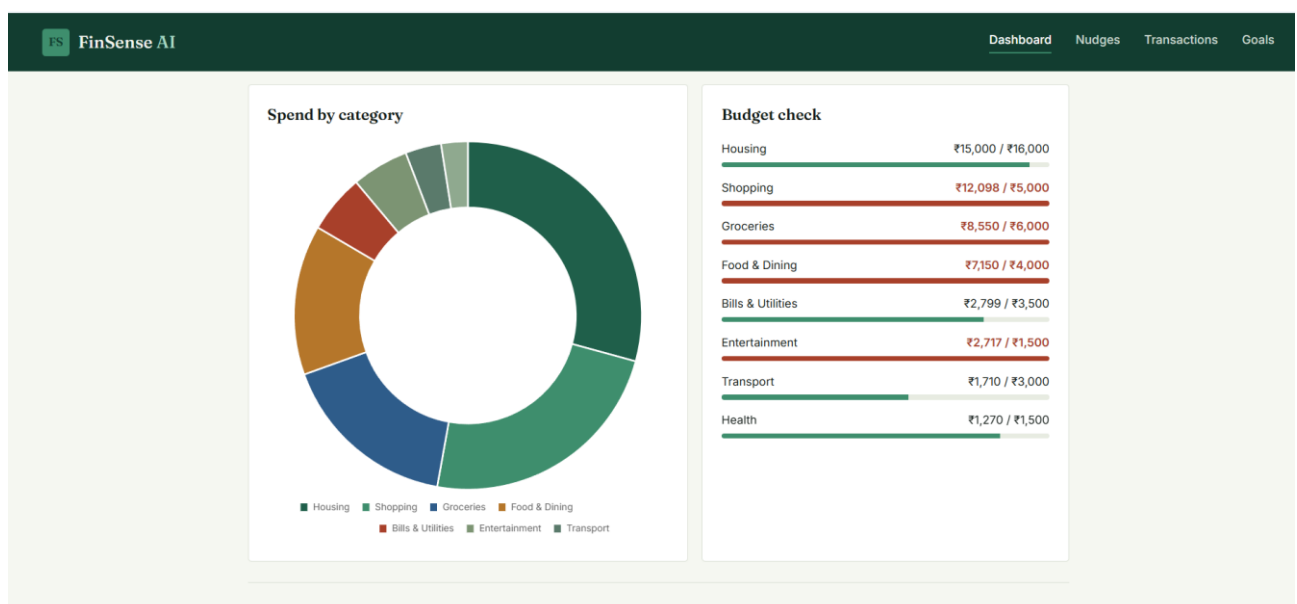


Figure 4.2: Transaction Categorisation

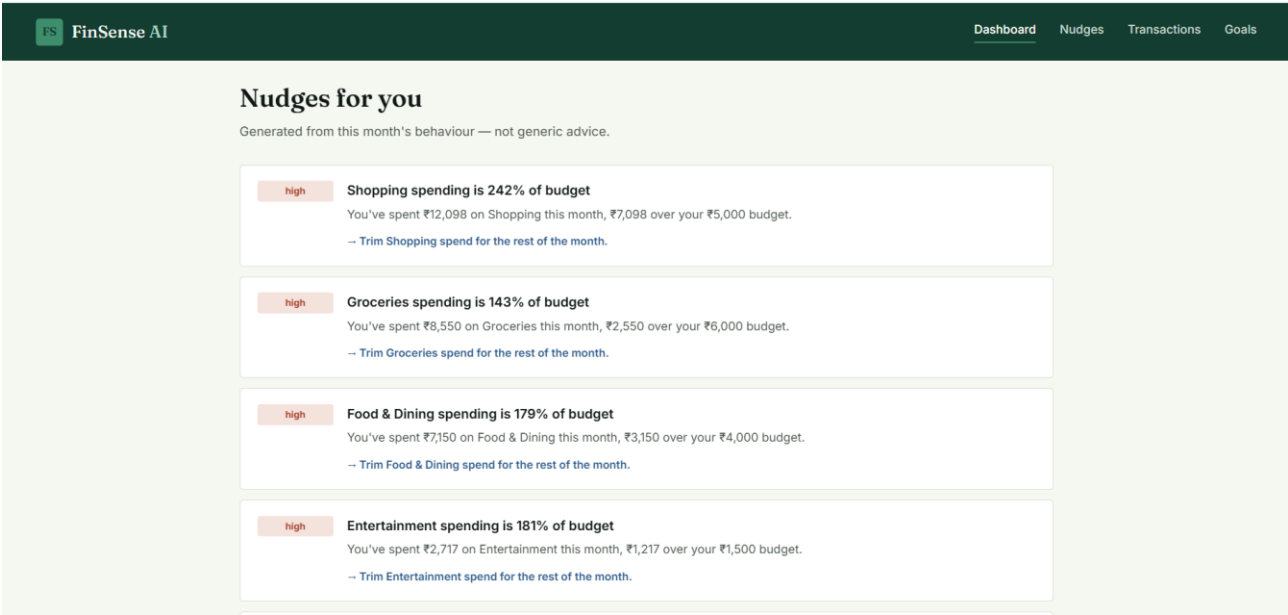


Figure 4.3: Spending Pattern Analysis

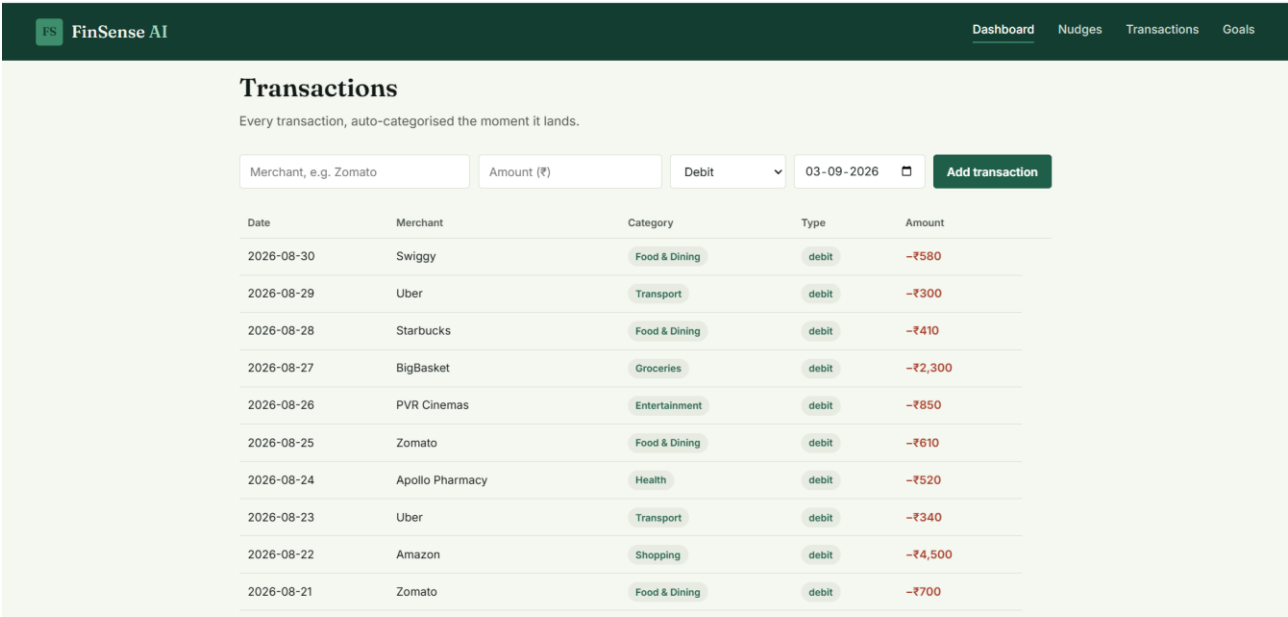


Figure 4.4: Overspending Detection

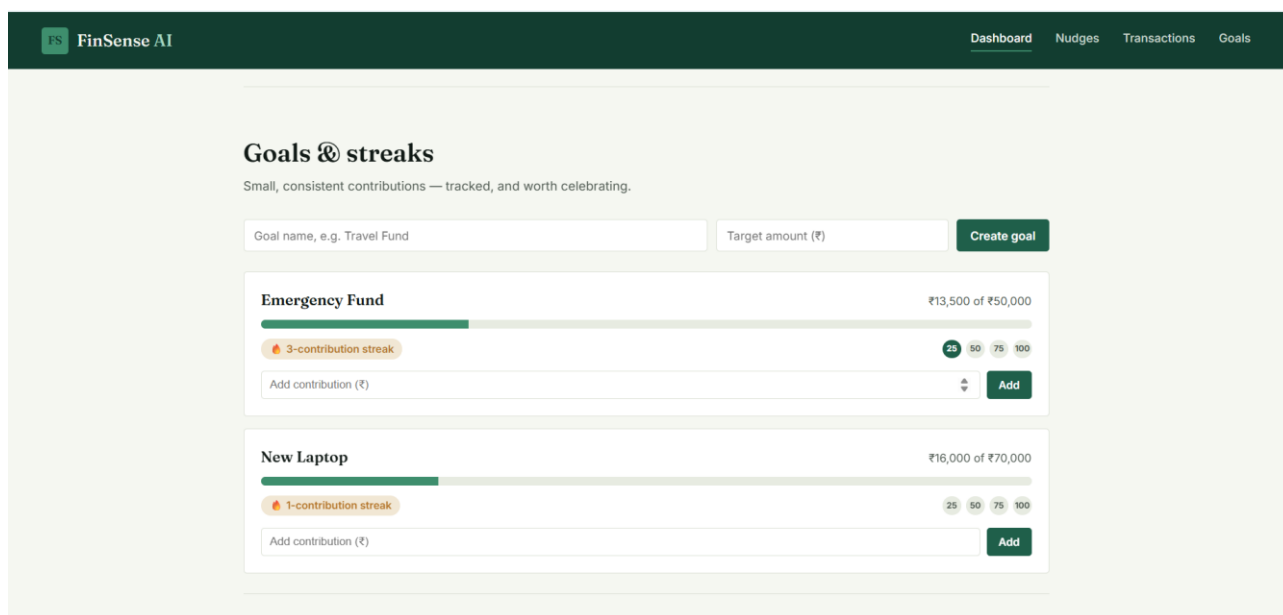


Figure 4.5: Spending Health Score

4.5 Data Analysis & Engineering Judgment

The transaction data is analysed to identify relationships between spending amount, category, frequency, and recurring behaviour.

A single high-value transaction may not always indicate poor financial behaviour. For this reason, the analysis considers spending patterns rather than only the total transaction amount. For example, a repeated series of small unnecessary expenses can become significant when viewed over a longer period.

The Spending Health Score is used to summarise the analysis. A stronger score indicates relatively controlled spending behaviour, while a lower score may indicate increased overspending, frequent unnecessary expenses, or unfavourable spending patterns.

The engineering judgment in Module 1 is that **continuous behavioural analysis is more useful than simply displaying a transaction history**. The analysed output can provide useful input for the later investment nudge and goal-tracking modules.

4.5 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Process transaction data	Transaction processing output	Fully Achieved
Categorise transactions	Category display	Fully Achieved
Analyse spending behaviour	Spending analysis	Fully Achieved
Detect overspending	Overspending results	Fully Achieved
Identify recurring expenses	Repeated transaction analysis	Fully Achieved
Identify unnecessary expenses	Behaviour analysis output	Fully Achieved
Generate Spending Health Score	Score output	Fully Achieved
Support other modules	Analysed spending data	Fully Achieved

Compared with basic expense tracking, the module provides an additional behavioural analysis layer.

4.7 Strengths & Limitations

Strengths

- Continuous transaction analysis.
- Automatic spending categorisation.
- Easy-to-understand spending information.
- Overspending identification.
- Recurring expense detection.
- Spending Health Score.
- Provides a foundation for personalised financial guidance.

Limitations

- Analysis quality depends on transaction data quality.
- New users may have limited historical data.
- Incorrect merchant information can affect categorisation.
- Spending behaviour can be complex and may not always be represented by a single score.

- The current academic implementation does not perform real banking transactions.
- The module is designed for financial guidance and does not replace

4.6 Project Planning, Roles & Budget

Project Milestone Plan

Milestone	Activity	Status
1	Problem identification	Completed
2	Requirement analysis	Completed
3	System design	Completed
4	Dataset preparation	Completed
5	Module 1 development	Completed
6	Module testing	Completed
7	Integration with other modules	Completed
8	Documentation	Completed

Individual Role

Student	Assigned Responsibility	Actual Contribution
Yuvasri S	Module 1 – Real-Time Spending Analysis Engine	Transaction analysis, categorisation, overspending detection, recurring pattern analysis and report preparation

Budget

Item	Estimated Cost	Actual Cost
Development Software	₹0 / Academic tools	₹0

Dataset / Sample Data	₹0	₹0
Development Environment	₹0	₹0
Internet and Testing Resources	Minimal	Minimal
Total	Minimal	Minimal

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **Real-Time Spending Analysis Engine** was developed as Module 1 of the FinSense AI project. The module addresses the problem of users being unable to clearly understand their spending behaviour from raw transaction records.

The implemented system processes transaction-level information and performs transaction categorisation, spending pattern analysis, overspending detection, recurring expense identification, unnecessary-expense identification, and Spending Health Score generation.

The module provides a strong analytical foundation for the complete FinSense AI system. The analysed spending behaviour can support the AI Investment Nudge Engine in identifying savings opportunities and can also support the Goal Tracking and Gamified Savings module.

The overall contribution of Module 1 is the transformation of basic transaction information into meaningful spending behaviour insights.

5.2 Future Work

The future development of Module 1 can include:

- More advanced personalised machine learning models.
- Improved automatic transaction categorisation.
- Improved spending prediction.

- Connection with additional financial data sources.
- Improved anomaly detection.
- Adaptive user-specific spending thresholds.
- More detailed Spending Health Score calculation.
- Enhanced privacy and security mechanisms.
- Natural-language explanations of spending patterns.
- Integration with real-time financial data sources.

These future directions are consistent with the project presentation's proposed future scope of advanced ML models, improved spending prediction, additional financial data sources, enhanced privacy and security, and improved behavioural prediction.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

Through the development of Module 1, I gained knowledge about:

- Financial transaction data processing.
- Spending behaviour analysis.
- Transaction categorisation.
- Overspending detection.
- Recurring transaction analysis.
- Data preprocessing.
- Financial data visualisation.
- AI/ML applications in personal finance.

Engineering and Professional Skills Developed

- Problem analysis.
- Software module design.
- Data analysis.
- Testing and debugging.
- System integration.
- Technical documentation.
- Teamwork.
- Project planning.

Individual Reflection – Yuvasri S

My main responsibility in this project was **Module 1 – Real-Time Spending Analysis**

Engine. The most difficult part was converting raw transaction information into meaningful spending behaviour instead of simply displaying transaction records. I worked on analysing transaction categories, spending frequency, overspending, recurring expenses, and the overall Spending Health Score.

A key engineering decision was to use a structured transaction-processing flow so that each stage produces information required by the next stage. This made the module easier to analyse and integrate with the remaining FinSense AI modules.

I gained new knowledge in transaction data processing, financial behaviour analysis, and the application of AI concepts to personal finance. If I repeat the project, I would improve the system using more personalised machine learning models and larger real-world transaction datasets.

REFERENCES

1. R. H. Thaler and C. R. Sunstein, *Nudge: Improving Decisions About Health, Wealth, and Happiness*. Yale University Press, 2008.
2. D. Kahneman, *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.
3. B. M. Barber and T. Odean, “The behavior of individual investors,” in *Handbook of the Economics of Finance*, vol. 2, Elsevier, pp. 1533–1570, 2013.
4. P. Sironi, *FinTech Innovation: From Robo-Advisors to Goal Based Investing and Gamification*. Wiley, 2016.
5. P. Gomber, J.-A. Koch, and M. Siering, “Digital Finance and FinTech: Current research and future research directions,” *Journal of Business Economics*, vol. 87, pp. 537–580, 2017.
6. J. Hamari, J. Koivisto, and H. Sarsa, “Does gamification work?—A literature review of empirical studies,” in *2014 47th Hawaii International Conference on System Sciences*, pp. 3025–3034, 2014.
7. Project Presentation: **FinSense AI – Real-Time Spending Behaviour Investment Nudge Engine**, Module 1 – Real-Time Spending Analysis Engine.